MISHESH PATEL

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PROFESSIONAL SUMMARY

B.Tech Computer Science & Engineering (Artificial Intelligence) student (expected May 2027) with a Python and algorithms foundation in AI application development. Built an AI-powered resume analysis system with the OpenAI API (GPT-4o-mini) and a weighted matching algorithm for personalized job recommendations, backed by REST APIs, data processing pipelines, and MongoDB/PostgreSQL databases. Seeking an entry-level AI/ML Developer, AI Engineer, or Applied AI role.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, SQL

AI / Applied AI: OpenAI API, GPT-4o-mini, AI Application Development, AI-Powered Resume Analysis, Weighted Matching Algorithm, Personalized Job Recommendations

Algorithms & Computer Science: Data Structures (Arrays, Linked Lists, Trees, Graphs), Algorithms (Sorting & Searching, Recursion, Dynamic Programming), Algorithm Design, OOP, Software Design Patterns

Data Processing: PDF Parsing (pdf-parse), File Upload Handling (Multer), MongoDB Aggregation Pipelines, SQL Migrations, Input Validation (Zod, express-validator)

Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST API Design, JWT Authentication, Role-Based Access Control (RBAC)

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Database Design (Schema Design, Indexing)

Tools & Testing: Git, GitHub, Jest, Supertest, React.js, HTML5, CSS3, Tailwind CSS, Vercel

PROJECTS

Career Gap Analyzer — AI-Powered Job-Matching Platform | *React*, *TypeScript*, *Node.js*, *Express.js*, *MongoDB*, *OpenAI API (GPT-4o-mini)*

- Integrated OpenAI GPT-4o-mini with a Multer (PDF-only, 5 MB limit) and pdf-parse pipeline to analyze uploaded resumes and extract 5 data points: skills, gap duration, gap risk, resume score, and suggested roles
- Implemented a weighted matching algorithm (60% skill match, 30% gap compliance, 10% resume score) with eligibility filters to generate personalized job recommendations for career-gap candidates
- Built a full-stack platform with React, TypeScript, Node.js, Express.js, and MongoDB supporting 3 user roles (student, company, admin) to match candidates with employers
- Secured backend REST APIs with JWT role-based route protection across 3 roles, bcrypt hashing (12 rounds), express-validator input validation, and 1 unique index preventing duplicate applications

Branded Short-Link & Bio-Link Hub — URL Shortener Platform | *React*, *TypeScript*, *Node.js*, *Express.js*, *MongoDB*

- Engineered a full-stack URL shortener and link-in-bio platform exposing 18 REST endpoints for links, analytics, authentication, and 5 customizable bio-page themes
- Designed a redirect algorithm with 6-character Base62 codes (56.8 billion combinations), unique-index collision handling (HTTP 409), and 302 redirects with asynchronous click logging so tracking never slows visitors
- Optimized analytics data processing with MongoDB aggregation pipelines and compound indexes across 4 collections, storing only HMAC SHA-256 hashed IPs
- Secured the API with JWT authentication (15-minute access, 7-day refresh tokens) in httpOnly cookies, token-family rotation, bcrypt hashing, Zod validation, and 4 rate limiters; validated with 29 Jest/Supertest tests across 5 suites (100% passing)

Mini ERP + CRM Operations Portal — Inventory & Sales Management | *React*, *TypeScript*, *Node.js*, *Express.js*, *PostgreSQL*

- Developed a full-stack ERP/CRM for wholesale distribution with 5 modules (auth, customers, products, stock movements, challans) and 19 REST endpoints
- Implemented RBAC for 4 user roles (Admin, Sales, Warehouse, Accounts) with JWT authentication (8-hour sessions), Zod request validation, and login rate limiting at 10 requests/minute per IP
- Engineered stock integrity by validating challan line items, then deducting stock and logging movements in 1 atomic PostgreSQL transaction with row-level locking (FOR UPDATE) to prevent race conditions
- Designed a relational schema with SQL migrations, custom enums, and indexed tables, applying 5 business rules (price snapshots, immutable stock log); added Jest/Supertest integration tests; deployed front end on Vercel

EDUCATION

Bachelor of Technology (B.Tech), Computer Science & Engineering (Artificial Intelligence) Expected May 2027

Parul Institute of Engineering and Technology, Vadodara, India CGPA: 7.33/10 | Merit Scholarship Recipient